DISCRIMINATION IN A NEURAL NETWORK'S SOCIETY

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ABSTRACT

Populations of learning agents are increasingly deployed as societies: language-model agents that read each other's outputs, weigh how far to trust them, and revise their positions accordingly. We ask whether discriminatory group structure can emerge in such a population from the *learning rule alone*, with no biased data and no group-level preference anywhere in the system. We study a society of adaptive agents that each carry a classifier and update, online and optimally, both their opinions and the trust they assign to every other agent, and inject the minimal possible bias: a fraction f_d of agents extend their representation of a message with a feature that carries no information about its content and depends only on the sender's group label, at strength d. The result is a phase diagram in (d, f_d) with four regimes separated by sharp transitions — neutral polarisation unrelated to group membership, two discriminatory phases distinguished by whether hostility still needs disagreement to sustain it, and, for bias favouring the out-group, a frustrated spin-glass regime in which no coherent faction can form. Discrimination requires a quorum of biased agents, and past it the transition is abrupt, so a population can be discriminatory while every agent in it passes an individual audit. We further show that the *order* of polarisation is controlled by the complexity of the agenda: on few issues distrust forms first and drags opinions after it, on many issues opinions separate first. Because the order parameters are defined only through pairwise opinion alignment and pairwise trust, they transfer directly to a society of language-model agents under comparable instructions, which we measure [results sentence], finding [agreement or disagreement with the predicted phase structure].

1 INTRODUCTION

Systems built from many language-model agents are now routine: agents that debate, review each other's work, negotiate, and simulate populations (Park et al., 2023; Du et al., 2024; Chuang et al., 2024). Each agent reads what others produce, forms some judgement of how much to rely on them, and adjusts its own position. That is a society, and societies have collective states that no individual chose. The question this paper asks is whether one of the collective states available to such a system is *discrimination*: a population that sorts itself into mutually distrustful groups along a label that carries no information.

The usual explanation for group bias in machine learning locates it in the data: a model trained on a biased corpus reproduces the bias. That account cannot explain a population-level effect, because it says nothing about interaction. We therefore ask a different question. Suppose every agent is individually well-specified — learning by an algorithm that is provably optimal in the setting it was designed for — and suppose the only defect is that some agents attend to one irrelevant feature of a message, namely who sent it. What does the *population* do?

We study this in a model in which the analysis can be carried out: a society of N agents, each carrying a perceptron over a K-dimensional embedding of issues, interacting by stating opinions about issues drawn from a shared agenda of size P. Two features make the model unusually apt for the question. First, agents can agree on some issues and disagree on others, so ideological distance is graded rather than binary. Second, learning is derived rather than posited: the update follows from a Bayesian step on the agent's belief, projected back onto a Gaussian family by maximum entropy (Opper, 1996; 1998; Caticha, 2020), and it carries alongside the weights an explicit, dynamical estimate of how noisy each other agent is as a channel — a quantity we can read as trust (Alves &

Caticha, 2016; Caticha & Vicente, 2011). Cognitive-sounding behaviour then appears without being written in: surprise, blame attribution between belief-revision and source-derision, and the ability to learn the *opposite* of what a distrusted source says.

Into this society we inject the smallest bias that can be correlated with a group label. A fraction f_d of agents shift their perceived agreement with a message by $\pm d$ according to whether the sender shares their label. Nothing else distinguishes the groups; the labels are informationally inert; the two groups are treated symmetrically, so any asymmetry that emerges is spontaneous. The resulting collective behaviour, as a function of (d, f_d), is the subject of the paper.

Contributions.

- **A phase diagram of emergent discrimination** (§6): four regimes with sharp boundaries — neutral polarisation unrelated to class, two discriminatory phases distinguished by whether ideology still tracks the group split, and a frustrated spin-glass regime when the bias favours the out-group. Discrimination needs a quorum: below $f_d \approx 0.4$ the society stays neutral however strong the bias, and above it the transition in d sharpens rapidly.
- **A mechanism specific to good learners** (§4): the bias displaces the separatrix of the learning flow, enlarging one dissonance-free basin at the other's expense. It acts only through the algorithm's trust sector, so optimising for the small-group setting is what makes a population vulnerable in the large-group one.
- **Which comes first, distrust or disagreement** (§6): the order of polarisation is set by the agenda complexity $\alpha = P/K$, reversing at $\alpha \approx 1.7$. This turns a contested question about polarisation (Fiorina & Abrams, 2008; Iyengar et al., 2012) into a statement about a measurable parameter.
- **The same measurements in a society of language models** (§7). The order parameters need only pairwise opinion alignment and pairwise trust, both elicitable from language agents, so the phase structure is testable on a substrate the theory does not describe. [one sentence of findings once the study is run]

Our reading of the theoretical result is deliberately narrow: these are simple machines, and extending conclusions from them to human societies is not automatic. What the model does support is a claim about learning systems — that an inference rule optimal for a pair of agents, transplanted into a large population, can convert an informationally worthless label into a stable architecture of group distrust. For systems assembled from many interacting learned agents that is a failure mode of the interaction rather than of the training data, and it is invisible to any evaluation that examines agents one at a time.

2 Related work

Societies of language-model agents. Multi-agent language systems have been used to simulate populations and to study what emerges from their interaction: generative agents with memory and social behaviour (Park et al., 2022; 2023), debate procedures that improve factuality (Du et al., 2024), opinion exchange over networks (Chuang et al., 2024), and language models as stand-ins for survey respondents (Argyle et al., 2023; Aher et al., 2023), with systematic biases documented in simulated debate (Taubenfeld et al., 2024). This literature is largely empirical: it reports what a particular population does. What it lacks is a model in which the collective outcome can be derived and its phase boundaries located. We supply one and treat the language-agent society as the system to be tested against it.

Bias and fairness in machine learning. The dominant framing locates group bias in data and in single-model decisions, as representational harm (Blodgett et al., 2020; Bender et al., 2021) or as allocative unfairness with formal criteria attached (Dwork et al., 2012; Hardt et al., 2016; Barocas et al., 2023). Our mechanism is orthogonal and complementary: the outcome here is produced by an irrelevant feature interacting with the *learning rule* across a group of agents, and it is undetectable by any audit that inspects agents one at a time, since a single agent's bias is $O(d)$ while the collective state it induces is discontinuous in d. That irrelevant features degrade a classifier is old; that they can restructure a *society* of classifiers is the point here.

Opinion dynamics and social balance. Models of consensus and polarisation in interacting populations run from bounded-confidence and mixing rules (Deffuant et al., 2000; Hegselmann & Krause, 2002) through culture dissemination (Axelrod, 1997) to reinforcement-driven echo chambers (Baumann et al., 2020); see Castellano et al. (2009) for a survey. In most of that work an agent's state is a scalar or a discrete label, so two agents either agree or do not. Representing agents as classifiers over an issue space (Caticha & Vicente, 2011) makes partial agreement possible, which is what lets us ask whether distrust follows disagreement or the reverse. Our second diagnostic, social balance over triples (Heider, 1946; Cartwright & Harary, 1956; Antal et al., 2005; Marvel et al., 2011), is usually applied to sign structures that are themselves the primitive objects; here both sign structures are generated by inference, and their balance is what separates organised bloc formation from disorder — the distinction that identifies the frustrated phase, which pair correlations alone would report as nothing happening.

Optimal on-line learning, and affective polarisation. The learning rule is the optimised on-line algorithm for the student–teacher problem, given a Bayesian reading by Opper (1996) and an entropic-dynamics formulation by Caticha (2020); the identification of the inferred channel-noise level with distrust, and its promotion to a dynamical variable, is due to Alves & Caticha (2016). Appendix B gives the fuller genealogy. Our contribution is not the algorithm but what it does to a population when one class-correlated feature is added. Separately, whether groups come to dislike each other because they disagree or the reverse is contested in political science (Fiorina & Abrams, 2008; Iyengar et al., 2012; Iyengar & Westwood, 2015); we do not adjudicate it, but identify a parameter that selects which occurs.

3 A SOCIETY OF AGENTS THAT LEARN FROM EACH OTHER

3.1 AGENTS, ISSUES AND OPINIONS

A society of N agents discusses an agenda of P issues. An issue is a vector $\hat{x} \in \mathbb{R}^K$: the embedding of an assertion in the K dimensions that are relevant to the question at hand. All agents embed issues identically and differ only in what they make of them, so the model separates the representation of an issue from the opinion an agent holds about it.

Each agent carries a single-layer perceptron with weights $w_I \in \mathbb{R}^K$ and states the opinion $\sigma_I = \mathrm{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. This is the least structure that lets two agents agree on some issues and disagree on others; a model in which each agent is a single spin cannot do that, and the possibility of partial agreement turns out to drive the collective behaviour we report. Agents also belong to one of two classes, A or B, recorded as $\kappa_I = \pm 1$: a label that is visible to others and carries no information about any issue.

An agent's state has two sectors. The *ideological* sector is a Gaussian belief over its own weights, with mean $\hat{w}_I$ and covariance C_I. The *affective* sector is a Gaussian belief, for every other agent J, over how unreliable J is as a source: mean $\mu_{J|I}$ (the *distrust* I assigns to J) and variance $V_{J|I}$. Writing the joint belief as a product of Gaussians,

$$Q(w_I, z_I \mid \lambda_I) \;=\; \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \tag{1}$$

the dynamical variables of the society are $\lambda_I = \{\hat{w}_I, C_I, \mu_{J|I}, V_{J|I}\}$. The means fix what an agent thinks; the (co)variances fix how firmly, and thereby how fast it can be moved.

3.2 ONE INTERACTION

At each step an issue $\hat{x}$, an emitter e and a receiver r are drawn uniformly. The emitter states σ_e; the receiver treats it as a noisy report of an unknown true label, where the noise level is precisely what its distrust of the emitter encodes. A Bayesian step on equation 1 leaves the exponential family, and projecting the posterior back onto a Gaussian by maximum entropy — relative to the prior, subject to the first two moments — keeps the belief in the form equation 1 and yields a closed update. This is the entropic-dynamics construction, and it is what makes the algorithm optimal in the small-group (student–teacher) setting rather than merely plausible. Appendix A carries the derivation, including the two modelling choices that keep the algebra Gaussian: taking the probability that the emitter conveys the wrong label to be $\Phi(z)$ with z Gaussian, and letting an inexperienced agent be correspondingly unsure it has parsed the issue as the emitter did.

Two scaled fields carry everything the update depends on:

$$h_w \;=\; \frac{\hat{w}_r \cdot \hat{x}\,\sigma_e}{\gamma_C}, \qquad h_\mu \;=\; \frac{\mu_{e|r}}{\gamma_V}, \qquad \gamma_C = \sqrt{1 + \hat{x}\cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \tag{2}$$

We call h_w the *opinion field*: it is positive when the receiver already agrees with the message and large when it holds that opinion confidently. We call h_μ the *distrust field*: positive when the receiver distrusts the emitter. In these variables the evidence for the message is

$$Z \;=\; \Phi(h_w) + \Phi(h_\mu) - 2\,\Phi(h_w)\Phi(h_\mu), \tag{3}$$

which is small in two quadrants — agreeing with a distrusted emitter, and disagreeing with a trusted one — and large in the other two. Low evidence is surprise, and surprise is what drives learning. The update is a gradient step in $\log Z$ in both sectors,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C}\,\sigma_e\, C_r \hat{x}, \qquad\qquad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2}\,(C_r\hat{x})(C_r\hat{x})^\top, \tag{4}$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V}\, V_{e|r}, \qquad\qquad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2}\, V_{e|r}^2, \tag{5}$$

with the four modulation functions given by the derivatives of $\log Z$:

$$\begin{aligned} F_w &= \big(1 - 2\Phi(h_\mu)\big)\frac{g(h_w)}{Z}, \qquad & F_\mu &= \big(1 - 2\Phi(h_w)\big)\frac{g(h_\mu)}{Z}, \\ F_C &= -F_w(F_w + h_w), & F_V &= -F_\mu(F_\mu + h_\mu), \end{aligned} \tag{6}$$

where g is the standard normal density. The first update in equation 4 is a Hebbian step $\sigma_e \hat{x}$ with a tensorial, self-annealing learning rate; F_C and F_V shrink the two uncertainties, so an agent's opinions and its judgements of others both harden with experience.

Three properties of equation 6 do the work in what follows, and none of them was put in by hand. **Trust gates learning and can invert it**: the prefactor $1 - 2\Phi(h_\mu)$ in F_w is negative when the emitter is distrusted, so a distrusted emitter is not ignored but *anti-learned* from, generating effectively antiferromagnetic couplings. (A comparable reversal is documented in humans, where a norm advertised by a distrusted source can push behaviour the other way, Keizer et al., 2011.) **Agreement builds trust**: symmetrically, $1 - 2\Phi(h_w)$ in F_μ is negative when the receiver agrees, so whether μ rises or falls is decided by the *sign of the perceived agreement* — the hinge that §4 turns. **Only one sector yields at a time**: the sectors are mirror images, $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$, and the surprise is absorbed by whichever field is smaller in magnitude, with a crossover at $h_\mu = h_w$ (Figure 1). Meeting a distrusted agent who agrees with you costs you either your distrust or your opinion, depending on which you hold less firmly — blame attribution emerging from the inference problem rather than being assumed.

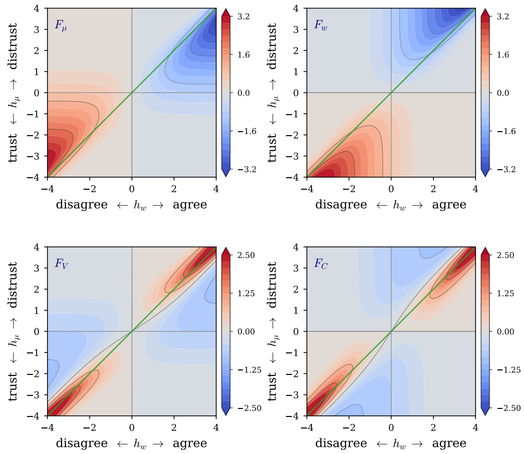
4 DISCRIMINATION AS AN ERROR OF REPRESENTATION

Nothing in §3 refers to an agent's class. We now let some agents make a specific, minimal mistake: when they receive a message, they extend the representation of the issue with a feature that carries no information about the issue at all, and depends only on *who sent it*. If that feature is correlated with the emitter's class label, the agent is a *discriminating* agent, and the resulting shift of its opinion field is the *discrimination field*:

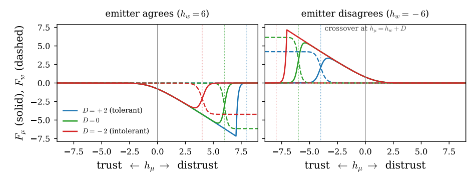
$$h_w^D \;=\; h_w + D_{\kappa_r,\kappa_e}, \tag{7}$$

where D is a 2×2 matrix indexed by the classes of receiver and emitter. A fraction f_d of the society discriminates, drawn independently of class; the rest use equation 2 unmodified.

This is a familiar failure viewed from an unusual angle: irrelevant features degrade a classifier, and if one correlates with a protected attribute its errors become structured along that attribute. Here the classifier is one agent among many, and the question is what those errors do to the *population*.



(a) The four modulation functions.



(b) Which sector absorbs the surprise.

Figure 1: **Learning is driven by dissonance.** (a) F_μ, F_w (top) and F_V, F_C (bottom) over the (h_w, h_μ) plane; the green line is $h_\mu = h_w$, and reflection across it exchanges the sectors. Changes concentrate in the dissonant quadrants: agree-and-distrust (upper right), disagree-and-trust (lower left). (b) Cuts at fixed h_w: below the crossover at $h_\mu = h_w + D$ the affective sector moves (solid), above it the ideological sector does (dashed) and the receiver unlearns. The discrimination field shifts that crossover, which is the whole mechanism studied here.

What the field does. By equation 6, the sign of perceived agreement decides whether trust grows or decays. A receiver that adds $+d$ to h_w reads its counterpart as more agreeable than it is and becomes more trusting: it is *tolerant*. A receiver that adds $-d$ manufactures disagreement that was not there and becomes more distrustful: it is *intolerant*. Throughout this paper $d > 0$ therefore denotes **tolerance towards in-group and intolerance towards out-group** emitters, and $d < 0$ the reverse — discrimination in favour of the other class.

Geometrically, the field displaces the separatrix of the learning flow to $h_\mu = h_w + D$ (Figure 4, Appendix C). The undisplaced flow has two symmetric consonant basins, "trust and agree" and "distrust and disagree", and a society whose agents fall into them polarises into two camps whose membership has nothing to do with class. Displacing the separatrix breaks that symmetry: a receiver meeting an out-group emitter has an enlarged "distrust and disagree" basin, and one meeting an in-group emitter an enlarged "trust and agree" basin. Every discriminating agent is thus biased towards the same alignment of distrust with class, and once enough of them are, the whole society can lock into it.

Six ways to discriminate. The matrix D can be filled in six inequivalent ways, listed in Appendix D: one class may favour its own without hostility to the other, be hostile without favouritism, or both; and either one class or both may discriminate. All the phase diagrams below use the fully symmetric case

$$D = d \begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}, \tag{8}$$

in which both classes favour their own and are hostile to the other. It is the case in which the two classes play equivalent roles, so any asymmetry that appears in the results is spontaneous rather than imposed.

A note on the sign convention. An earlier version of this model (Anonymous, 2026) tabulates D with the opposite sign while placing the discriminatory phases at $d > 0$ in its text and figures; the two are inconsistent, and the inconsistency exchanges the halves of the phase diagram. Appendix E works this through and shows the maps under both readings. We use equation 7 with equation 8 as stated, which is the reading that agrees with the algorithm.

5 WHAT TO MEASURE

A society's state is a set of weight vectors and an $N \times N$ matrix of distrust. To compare states — and, later, to compare a society of perceptrons with a society of language models — we need statistics that depend on neither the dimension K nor the parameterisation.

Two pairwise quantities do the work. The *ideological overlap* of two agents is the cosine of their weight vectors, $\rho_{IJ} = \hat{w}_I \cdot \hat{w}_J / (\|\hat{w}_I\| \|\hat{w}_J\|)$, which is $+1$ when they agree on every issue and -1 when they disagree on every one. The *trust* a receiver I places in an emitter J is $\eta_{J|I} = 1 - 2\Phi(h_\mu)$, which is $+1$ when J is fully trusted and -1 when fully distrusted: the distrust field mapped onto a bounded, sign-carrying scale. Writing $G_{IJ} = \kappa_I \kappa_J$ for the class indicator ($+1$ for same-class pairs, -1 otherwise), the three pair correlations are

$$R_{w,\mu} = \big\langle (\eta_{I|J} + \eta_{J|I})\, \rho_{IJ} \big\rangle, \qquad R_{\mu,c} = \big\langle G_{IJ} (\eta_{I|J} + \eta_{J|I}) \big\rangle, \qquad R_{c,w} = \big\langle G_{IJ}\, \rho_{IJ} \big\rangle, \tag{9}$$

averaged over unordered pairs and normalised to $[-1, 1]$. $R_{w,\mu}$ asks whether agents trust those who agree with them; $R_{\mu,c}$, the order parameter of discrimination, asks whether trust follows class; $R_{c,w}$ asks whether opinion follows class.

The pair correlations cannot distinguish a society split into two coherent camps from one in which distrust is spread inconsistently, so we also measure social balance in the sense of Heider (1946) and Cartwright & Harary (1956). A triple of agents is ideologically balanced when $b^I_{IJK} = \rho_{IJ}\rho_{JK}\rho_{KI} > 0$ and affectively balanced when $b^A_{IJK} = \frac{1}{2}(\eta_{IJ}\eta_{JK}\eta_{KI} + \eta_{JI}\eta_{IK}\eta_{KJ}) > 0$; the aggregates

$$B_I = \big\langle b^I_{IJK} \big\rangle_{(IJK)}, \qquad B_A = \big\langle b^A_{IJK} \big\rangle_{(IJK)} \tag{10}$$

run over all $\binom{N}{3}$ triples. A randomly initialised society has $B_I \approx B_A \approx 0$: half its triples are frustrated. A society that has settled into two internally coherent, mutually opposed blocs has $B_I = B_A = 1$, whatever the blocs are made of — and a society that cannot settle at all stays near zero or goes negative. Balance therefore separates *organised* disagreement from mere disorder, which is exactly the distinction the phase diagram turns on. Both aggregates are computed in closed form from matrix products rather than by enumerating triples (Appendix G).

Note that none of equation 9 or equation 10 mentions weights, gradients or K: they need only a signed opinion alignment and a signed trust per pair, both of which are elicitable from a language agent. That is what makes the phase diagram of §6 a prediction §7 can test rather than an analogy.

6 THE PHASE DIAGRAM

We simulate societies of $N = 40$ agents over a 200×200 grid in (d, f_d), one society per grid point, and measure the five order parameters of §5 after a fixed number of interactions, with both classes discriminating (case equation 8). Parameters and their provenance are in Appendix H; each pixel is a single realisation, so the speckle is society-to-society fluctuation, not a rendering artefact.

Discrimination needs a quorum, and then arrives abruptly. At small f_d the society stays neutral no matter how strong the bias: the few discriminating agents are outvoted by the class-blind majority they learn from. The trust–class correlation at large d first exceeds 0.5 around $f_d \approx 0.4$ (0.43 for the simple agenda, 0.35 for the complex one), and past that the transition in d sharpens quickly: at $f_d = 1$ the band in which $|R_{\mu,c}| < 0.25$ is only 0.54 wide, and $R_{\mu,c}$ runs from -0.99 at $d = -1$ to $+0.99$ at $d = +1$. This is the first practical consequence of the model: the state of the population is not a smooth function of how biased its members are, and a minority of biased agents leaves no trace at all until it reaches a threshold fraction.

Two discriminatory phases, and only the simple agenda has both. Both discriminatory regions have distrust aligned to class; they differ in whether opinion is aligned too. Along $f_d = 0.9$ in the top row of Figure 2, $R_{c,w}$ rises to 0.44 at $d \approx 0.6$ and then *falls* to 0.30 by $d = 1$, while $R_{\mu,c}$ keeps climbing, from 0.78 to 0.87 — a gap of 0.55 where the two started level. Ideological balance follows $R_{c,w}$ down, B_I dropping from 0.19 to 0.06 across the same row. At moderate bias the society splits into two camps that both distrust and disagree with each other (region III); at

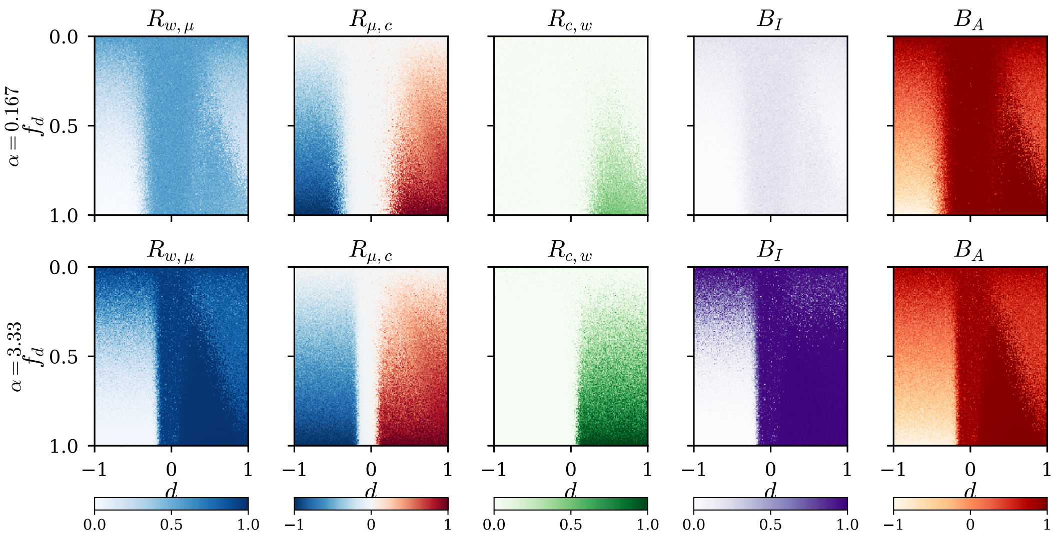


Figure 2: **All five order parameters over the** (d, f_d) **plane**, for a simple agenda ($\alpha = 0.17$, top) and a complex one ($\alpha = 3.3$, bottom). $R_{\mu,c}$ is the discrimination order parameter, flipping sign at $d \approx 0$ ever more sharply as f_d grows; $R_{c,w}$ is non-zero only for $d > 0$ and, for the simple agenda, *falls back* at the largest d, which is what separates the two discriminatory phases; B_I separates the agenda sizes, featureless above and saturated below; and B_A is the only panel in which the frustrated region at lower left announces itself, going strongly negative where everything else merely goes to zero.

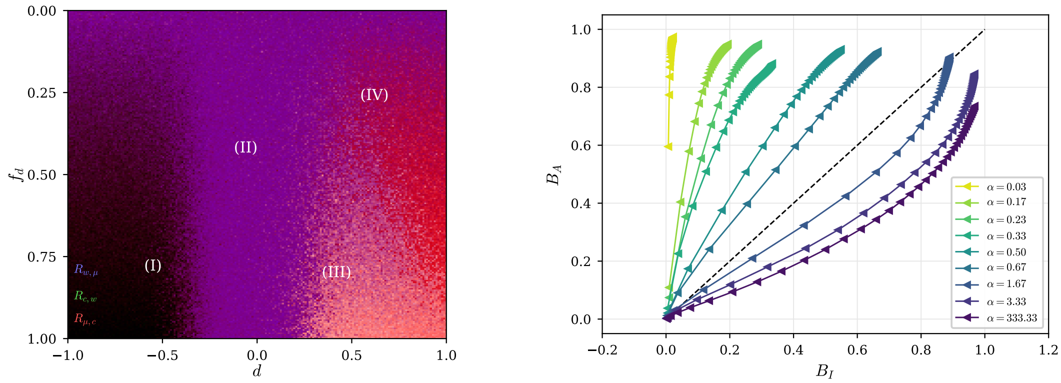
strong bias the disagreement becomes unnecessary and class membership alone carries the hostility (region IV), which is what the magenta corner of Figure 3a shows.

The complex agenda has no region IV. In the bottom row, $R_{\mu,c}$ and $R_{c,w}$ track each other across the whole $d > 0$ half — 0.84 against 0.85 at $d = 0.15$, 0.91 against 0.91 at $d = 0.75$, and still only 0.04 apart at $d = 1$ against the simple agenda's 0.55 — while B_I saturates at 0.98. When there are many issues to disagree about, the ideological split tracks the class split completely, so distrust and disagreement never come apart. Class-only hostility is available only to a society whose agenda is too narrow for ideology to reorganise around the class boundary.

Favouring the out-group produces frustration, not inverted discrimination. For $d < 0$ the correlations are the sign-mirror of the $d > 0$ side, which invites reading the left half of the diagram as "reverse discrimination". The balance measures say otherwise (Figure 2). In the discriminatory region $B_A = +0.97$: nearly every triple is affectively balanced, as two coherent mutually hostile blocs require. At $d = -1$, $f_d = 0.9$, $B_A = -0.71$ for both agendas — strongly *anti*-balanced — with $B_I = 0.00$. No consistent assignment of trust exists: each agent is pulled towards the other class while its own neighbours are pulled towards theirs, and the society settles into a spin glass rather than into a mirror-image society. This is also the one place where $R_{w,\mu}$ collapses, to 0.03. Pair correlations alone would report the region as simply inverted; the triple-level measurement identifies it as disordered.

Which polarisation comes first depends on the agenda. With no discrimination at all, a society still polarises. Figure 3b follows its path through the (B_I, B_A) plane from the random initial condition, where half the triples are frustrated, towards the frustration-free corner. For a simple agenda the path runs up the B_A axis: affective structure forms almost completely before ideological structure begins, ending at $(B_I, B_A) = (0.02, 0.97)$ for $\alpha = 0.03$. For a complex agenda the path stays below the diagonal, ending at $(0.97, 0.74)$ for $\alpha = 333$: opinions separate first and distrust follows. The crossover is at $\alpha \approx 1.7$, where the trajectory lies along the diagonal — that is, when the number of issues in play is comparable to the dimension of the space they are represented in.

Read as a mechanism, this says that the question of whether groups come to dislike each other because they disagree, or come to disagree because they dislike each other, has a parameter attached



(a) The four phases.

(b) Which polarisation comes first.

Figure 3: **(a) Four collective states.** The three correlations of Figure 2 (simple agenda) composited as colour channels — red $R_{\mu,c}$, green $R_{c,w}$, blue $R_{w,\mu}$ — so colour names the phase: (I) dark, frustrated reverse discrimination; (II) blue-violet, neutral polarisation unrelated to class; (III) pale, ideology tracking class; (IV) magenta, class alone. **(b) Balance trajectories** at $d = 0$, one curve per agenda complexity, markers equally spaced in time (they crowd at the end because the annealed dynamics slows). Simple agendas run above the diagonal, complex ones below, crossing over at $\alpha \approx 1.7$.

to it rather than an answer. Where the agenda is narrow — few issues, functioning as symbols — affect leads. Where it is broad, ideology leads. The same parameter explains why region IV exists only in the top row of Figure 2: with a simple agenda and a strong field, ideology has not had time to reorganise around the class boundary, so hostility arrives without it.

7 A SOCIETY OF LANGUAGE-MODEL AGENTS

The phase diagram of §6 is a claim about agents that learn from one another under a trust-weighted rule. Perceptrons instantiate that rule exactly; language-model agents instantiate something we cannot write down. If the phase structure survives the substitution, it is a property of the interaction rather than of the parameterisation.

N language-model agents share an agenda of P issues and interact in rounds mirroring §3: an issue, an emitter and a receiver are drawn; the emitter states its position with a brief justification; the receiver, seeing the statement and the emitter's group label, updates both its own position on that issue and its disposition towards that emitter. Nothing instructs an agent to agree with, trust, or favour anybody. The two axes of the phase diagram become two knobs on the instructions: f_d is the fraction of agents whose prompt carries a discriminating clause, assigned independently of group, and d is the strength of that clause — a graded instruction to treat the other group's statements as less credible than their content warrants and one's own group's as more credible, at several levels including $d = 0$ (clause absent) and $d < 0$ (the clause reversed). Group labels name arbitrary partitions with no content relating to any issue, so that class-aligned structure has to be produced by the interaction rather than retrieved from what the model already associates with a real social category. Appendix I gives the protocol in full.

Measurement needs only the two quantities of §5, both elicited outside the interaction loop so that measuring does not perturb the dynamics. Each agent rates all P issues on $[-1, 1]$, giving an opinion vector v_I whose normalised inner products are the overlaps ρ_{IJ}; and each agent reports, for every other agent, how far it trusts that agent's statements, mapped to $\eta_{J|I} \in [-1, 1]$ and deliberately not symmetrised. The three correlations equation 9 and the two balance aggregates equation 10 then follow unmodified, which is the point of having defined them without reference to weights or to K.

7.1 RESULTS

[To be written from the run. The comparisons it must make, in order of importance: (i) the sign and magnitude of $R_{\mu,c}$ against d at fixed f_d, versus the sharp transition at $d \approx 0$ of Figure 2; (ii) whether the two discriminatory regimes are distinguishable, i.e. whether $R_{c,w}$ falls while $R_{\mu,c}$ stays high as d grows; (iii) whether $d < 0$ gives the frustrated signature $B_A < 0$ with $B_I \approx 0$ rather than simple reverse discrimination; (iv) the agenda-complexity reversal.]

A match would establish the phase structure as a property of trust-weighted social learning as such — and, the transition being sharp, that a small change in how agents are told to treat each other can move a deployed population between regimes. A mismatch is equally informative, and we commit in advance to reporting which of the four comparisons fails. The likely failure modes are statements about the substrate rather than the model: stated trust compressed towards the positive end, instruction drift making d non-stationary, or an agenda too small to reach $\alpha > 1$.

8 DISCUSSION

Optimality is context-dependent, and the context here is group size. The rule these agents use is optimal for the problem it was derived for: one student, one teacher, a noisy channel whose reliability must be inferred online. Nothing about it is careless. Yet the same rule, run in a large population with one class-correlated perturbation, converts an informationally empty label into a stable architecture of group distrust. The mechanism is specific: it acts through F_μ, and a rule with no affective sector has none to bias — under a plain Hebbian update the discrimination field does nothing at all, since the update does not depend on h_w.[the direct ablation is not run; see NOTES.md] What exposes the population is precisely the machinery that makes each agent good at inference: that it draws conclusions about *sources* as well as about the world. Giving interacting agents a well-founded model of each other's reliability is not a neutral improvement.

Individually small, collectively discontinuous. A single discriminating agent is biased by $O(d)$; the population's state is not, moving between phases over a narrow interval in d once f_d passes its threshold. An evaluation protocol that inspects agents one at a time — the standard protocol — can therefore certify every agent as approximately unbiased while the population sits in a discriminatory phase. Auditing a multi-agent system requires population-level order parameters, and the ones used here cost two elicited matrices per society.

Two results that reframe standard questions. The reversal at $\alpha \approx 1.7$ says that whether groups come to dislike each other because they disagree, or the reverse, is not a choice between rival theories but two parameter regimes of one system. And making d negative, so that agents favour the out-group, gives not a tidy inverted society but frustration: $B_A < 0$ with $B_I \approx 0$, order parameters near zero not because nothing is happening but because nothing can settle. Pair correlations alone would misreport that state as neutral; only the balance measures identify it as a spin glass.

Limitations. The agents are single-layer perceptrons over a fixed shared embedding, all pairs may interact, and the society is closed and of fixed size. The dynamics anneals rather than reaching a stationary state, so our phases are phases at a finite interaction count. The class labels are exactly inert, the clean version of a case that is never clean in practice. And the map from an instruction given to a language agent to a numerical d is a calibration, not an identity: §7 tests whether the structure transfers, not whether the two systems agree quantitatively. Extending any of this to human societies is a further step the model does not license.

REPRODUCIBILITY STATEMENT

Every figure in this paper is produced by a script in the accompanying code, which contains the model, the order parameters, and the sweep driver, together with a test suite that checks the modulation functions against finite differences of $\log Z$, the invariants of the dynamics (the covariance stays symmetric positive-definite, the affective variance stays positive), and the order parameters against their known limits. One command regenerates all figures at a choice of three resolutions; sweeps are cached, so a figure can be restyled without re-simulating.

The model has few parameters and we state all of them, including the ones the formulation leaves open. §6 gives the values used; Appendix H gives the full table, separating quantities fixed by the model from those we chose, and records how each choice was made and what happens under alternatives. Two conventions where a literal reading of the model as previously written is ambiguous or inconsistent — the sign of the discrimination field, and the normalisation of the class indicator — are resolved explicitly in Appendices E and F, with both readings implemented behind a flag so either can be reproduced.

For the language-agent study we report the model version and decoding parameters, the full text of every prompt, the elicitation protocol, the number of agents, rounds and repeats, and the raw elicited opinion and trust matrices. [fill in once the study is run; the transcripts and matrices are to be released with the code]

Ethics statement

This paper studies how discriminatory structure can arise among interacting learning agents. Two points bear stating.

The model is not a claim about people. Its agents are single-layer perceptrons, its groups are arbitrary inert labels, and its "tolerance" is a single scalar. We use the vocabulary of in-group and out-group because it names the structure compactly, not because the model licenses inferences about human prejudice, whose causes are historical, institutional and material in ways nothing here represents. Where we discuss the affective-versus-ideological polarisation literature, we do so to point out that the model identifies a parameter that selects between two regimes, not to adjudicate an empirical question about any actual polity.

The language-agent study instructs agents to treat one group's statements as less credible than another's, and measures the result. This is the intervention under study, so it cannot be avoided, but it is confined: the groups are arbitrary partitions with no reference to any real social category, protected or otherwise, and the labels carry no content connected to the agenda. No human subjects are involved and no personal data is used. We report the instruction wording in full (Appendix I) because reproducibility requires it and because readers should be able to judge how strong the intervention is.

The practical direction of the work is defensive. The result implies that a multi-agent system can occupy a discriminatory collective state while every agent in it passes an individual audit, and that the transition into that state is sharp in the strength of a per-agent bias. That argues for population-level measurement of deployed multi-agent systems, and the order parameters used here are inexpensive enough to serve as one.

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A Derivation of the learning algorithm

The receiver's belief before an interaction is the product of Gaussians equation 1. It observes σ_e, the emitter's opinion on issue $\hat{x}$, and treats it as a report of an unknown true label σ_T transmitted through a channel that flips the label with probability $\varepsilon(z)$. Marginalising the joint distribution of labels and of the issue as the emitter may have parsed it, y_t,

$$L(\sigma_e \mid \hat{x}, u) \; = \; \sum_{\sigma_T} \int P(\sigma_e \mid \sigma_T, y_t, \hat{x}, u)\, P(\sigma_T \mid y_t, \hat{x}, u)\, P(y_t \mid \hat{x}, u)\, \mathrm{d}y_t, \tag{11}$$

with $P(\sigma_e \mid \sigma_T, z) = (1-\varepsilon(z))\delta_{\sigma_e,\sigma_T} + \varepsilon(z)\delta_{\sigma_e,-\sigma_T}$ and $P(\sigma_T \mid y_t, w) = \Theta(\sigma_T\, y_t \cdot w)$. Two modelling choices keep the algebra Gaussian. Taking $\varepsilon(z) = \Phi(z)$ with z Gaussian of mean $\mu_{e|r}$ and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the affective sector — a Beta distribution for ε, the tempting alternative, does not. Taking $P(y_t \mid \hat{x}, u)$ Gaussian with variance $v_p = \|w\|^{-2}$ lets an inexperienced agent, whose weights are small, be correspondingly unsure that it has parsed the issue as the emitter did. Together these give the likelihood

$$L(\sigma_e \mid z, \hat{x}, w) \;=\; \Phi(z)\,\Phi(-w\cdot\hat{x}\sigma_e) + \big(1-\Phi(z)\big)\,\Phi(w\cdot\hat{x}\sigma_e), \tag{12}$$

and the covariance stays block diagonal in the two sectors if it starts so.

Averaging the likelihood over the current belief gives the evidence

$$Z(\sigma_e \mid \hat{x}, D_t) \;=\; \mathbb{E}_{Q_t}\big[L(\sigma_e \mid z, u, \hat{x})\big] \;=\; \sum_{\sigma_T} \mathbb{E}_z\big[P(\sigma_e \mid \sigma_T, \hat{x}, z)\big]\,\mathbb{E}_w\big[\Phi(\sigma_T\, w\cdot\hat{x})\big], \tag{13}$$

and the Gaussian integrals collapse onto the two scaled fields equation 2, giving equation 3: the evidence depends on the receiver's state only through (h_w, h_μ). This is the symmetry that makes the two sectors mirror images.

The exact Bayesian posterior is not Gaussian. Replacing it by the Gaussian of maximum entropy relative to the prior, subject to matching the first two moments of the posterior, gives updates in terms of derivatives of the log evidence,

$$\hat{w}_{t+1} = \hat{w}_t + C_t\frac{\partial \ln Z}{\partial \hat{w}}, \qquad C_{t+1} = C_t + C_t\frac{\partial^2 \ln Z}{\partial \hat{w}\,\partial \hat{w}^\top}C_t, \tag{14}$$

and identically in the affective sector with $\mu_{e|r}, V_{e|r}$ in place of $\hat{w}, C$. Expressing the derivatives in the scaled fields yields equation 4–equation 5 with the modulation functions equation 6. The first is a Hebbian term $\sigma_e\hat{x}$ with a tensorial, self-annealing rate; the rest is the online estimation of the channel.

B Genealogy of the learning rule

The update equation 4–equation 5 descends from the variational optimisation of modulated Hebbian on-line learning (Kinouchi & Caticha, 1992; Kinzel & Ruján, 1990), given a Bayesian reading by Opper (1996; 1998) and applied across architectures by Solla & Winther (1999); the entropic-dynamics formulation we use is Caticha (2020). Learning through a noisy channel whose noise level must itself be inferred was analysed by Kinouchi & Caticha (1993) and Biehl et al. (1995); Alves & Caticha (2016) identified that inferred level with distrust and made it dynamical, with applications to voting (Caticha et al., 2015) and to panels of appellate judges (Caticha & Alves, 2019), and a mean-field treatment by Simões & Caticha (2018). The blame-attribution behaviour resembles the least-action rule of Mitchison & Durbin (1989), which suffers retarded learning (Kabashima, 1994), whereas the optimised algorithm attains generalisation error decaying as α^{-1} even under output noise (Simonetti & Caticha, 1996).

C The learning flow under a discrimination field

Figure 4 shows the flow directly.

D The six discrimination cases

The matrix D in equation 7 admits six inequivalent fillings, where $+d$ is the tolerant entry and $-d$ the intolerant one (§4 and Appendix E). Rows index the receiver's class, columns the emitter's.

All phase diagrams in the main text use case 6, in which the two classes play symmetric roles. [optional: add the case-1 and case-4 phase diagrams here if space allows, to show that one-sided discrimination produces a weaker but qualitatively similar transition]

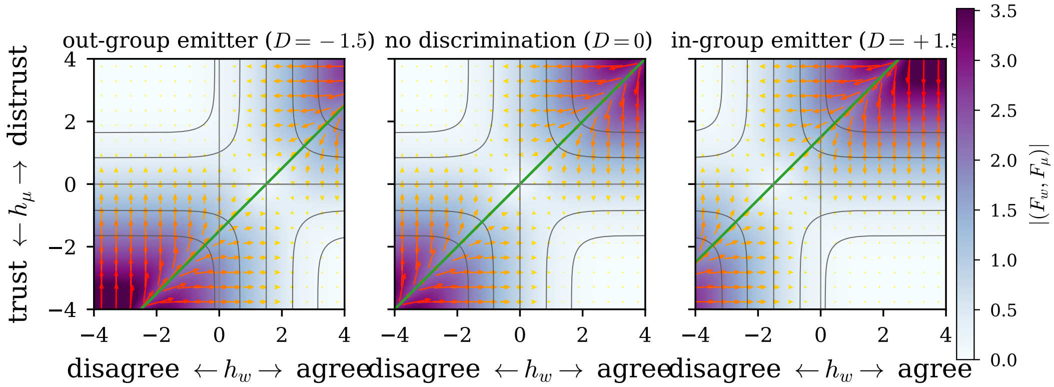

Figure 4: **The discrimination field bends the flow of learning.** Arrows are the change (F_w, F_μ) that one interaction induces in the receiver's fields; the background is $|(F_w, F_\mu)|$ and the thin contours are the evidence Z. Centre: without discrimination the flow escapes the two dissonant quadrants into two symmetric consonant basins, separated by $h_\mu = h_w$ (green). Left and right: a discriminating receiver evaluates the flow at $h_w + D$, which moves the separatrix and enlarges one basin at the other's expense — "distrust and disagree" for an out-group emitter ($D = -d$), "trust and agree" for an in-group one ($D = +d$).

case	D/d	who discriminates	how
1	$\begin{pmatrix} +1 & 0 \\ 0 & 0 \end{pmatrix}$	A only	in-group favouritism
2	$\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$	A only	out-group hostility
3	$\begin{pmatrix} +1 & -1 \\ 0 & 0 \end{pmatrix}$	A only	both
4	$\begin{pmatrix} +1 & 0 \\ 0 & +1 \end{pmatrix}$	both	in-group favouritism
5	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	both	out-group hostility
6	$\begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}$	both	both

E THE SIGN OF THE DISCRIMINATION FIELD

The sign of D is fixed by the algorithm and not free. By equation 6, $F_\mu = (1 - 2\Phi(h_w))g(h_\mu)/Z$ is negative when $h_w > 0$, so perceived agreement drives distrust μ *down*. A receiver that adds $+d$ to h_w therefore becomes more trusting (tolerant) and one that adds $-d$ becomes more distrustful (intolerant). Case 6 with $d > 0$ consequently gives in-group trust and out-group distrust, i.e. the discriminatory phase, which is the convention used throughout.

The companion manuscript (Anonymous, 2026) tabulates the in-group entry as $-d$ while its text and figures place the discriminatory phases at $d > 0$; those two statements are inconsistent, and taken literally the tabulated signs put the discriminatory phases at $d < 0$, mirroring every map in d. The same inconsistency appears in its description of the learning flow, where the panel labelled $D_{e|r} = -d$ is called the more tolerant one although $-d$ is the intolerant entry, and where the "distrust and disagree" basin is said to grow for $d > 0$ although adding $+d$ shrinks it. A single global sign reconciles everything; we adopt it in the form that leaves equation 7 untouched. The accompanying code implements both readings behind a flag, and Figure 5 runs the same sweep under each: the two rows are mirror images in d, and only the first matches the maps as published.

F CONVENTIONS IN THE ORDER PARAMETERS

Two choices in equation 9 deserve statement.

The class indicator. We use $G_{IJ} = \kappa_I \kappa_J \in \{-1, +1\}$. A $\{0, 1\}$ indicator, which counts only same-class pairs, cannot represent anti-correlation: under perfect reverse discrimination it returns $-(N/2 - 1)/(N - 1)$ rather than -1, and its range depends on N. Since the reverse-discrimination half of the phase diagram is a substantive part of the result, the signed indicator is the appropriate one. Both are implemented.

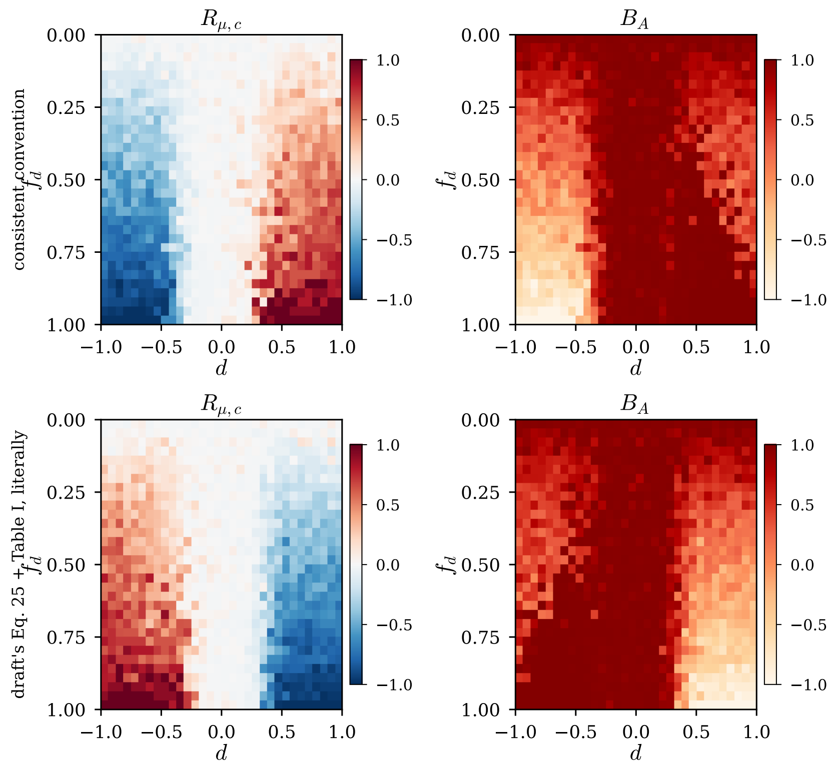

Figure 5: **The two readings of the discrimination field.** Trust–class correlation and affective balance over the (d, f_d) plane under the convention used in this paper (top) and under the companion manuscript's table read literally with its Eq. 25 (bottom). The physics is identical; which half of the d axis is discriminatory is not.

Normalisation. $R_{w,\mu}$ and $R_{\mu,c}$ sum $\eta_{I|J} + \eta_{J|I}$ over unordered pairs, so dividing by $N(N-1)$ places them in $[-1, 1]$. $R_{c,w}$ has a single term per pair and the same divisor would cap it at $1/2$; we use $2/(N(N-1))$ so that all three share the range $[-1, 1]$ and the three colour channels of Figure 3a are commensurate.

G Closed form for the balance aggregates

Enumerating triples costs $O(N^3)$ per society, which is prohibitive across a phase diagram. For any matrix M with unit diagonal — either ρ or η —

$$\sum_{\substack{(I,J,K)\\ \text{distinct, ordered}}} M_{IJ}M_{JK}M_{KI} \;=\; \mathrm{tr}(M^3) - 3\Big(\textstyle\sum_{I,J} M_{IJ}M_{JI} - N\Big) - N, \tag{15}$$

and each unordered triple appears six times, three as the forward cycle and three as the reverse, so $\langle b\rangle = \big[\text{right-hand side}\big]/\big(6\binom{N}{3}\big)$. For symmetric M this is B_I; for the asymmetric trust matrix the two cycle orientations are exactly the two terms of b^A, so the same expression gives B_A. Both are batched matrix products. The identity is verified against enumeration in the test suite.

H Parameters

The model needs six numbers. One is recovered from the companion manuscript, one is calibrated against its published figures, two are prior conventions, and two are free choices; all six are listed with their provenance rather than left implicit, since the manuscript states none of them. The accompanying code exposes each of them and its calibration script reproduces the comparisons below.

symbol	meaning	value	provenance
K	dimension of the issue embedding	30	recovered (see below)
N	agents in the society	40	chosen; converged (see below)
P	issues on the agenda	5 / 100	chosen: $\alpha = P/K$ either side of 1
Δt	interactions per ordered pair	500	calibrated (see below)
C_0, V_0	initial uncertainties	I, 1	uninformative prior
μ_0	initial distrust	$\mathcal{U}(-1, 1)$	gives $B_I \approx B_A \approx 0$ at $t = 0$

K is recovered, not chosen. The companion manuscript reports balance trajectories at $\alpha \in \{0.03, 0.17, 0.23, 0.33, 0.50, 0.67, 1.67, 3.33, 333.33\}$. These are exactly $P/30$ for $P \in \{1, 5, 7, 10, 15, 20, 50, 100, 10^4\}$, which fixes $K = 30$; the language-agent protocol's thirty issues agrees.

Δt is calibrated against the published trajectories. The dynamics anneals rather than reaching a stationary state, so the measurement time matters and the companion manuscript does not state it (its text carries a literal placeholder in place of the value). We scan Δt and compare the endpoints of all nine balance trajectories against the published ones:

Δt (interactions per ordered pair)	60	125	250	**500**	1000
rms distance to published endpoints	0.63	0.31	0.15	**0.13**	0.15

The minimum is flat-bottomed and sits at $\Delta t = 500$, which we use throughout. Seven of the nine endpoints then agree to within about 0.1, and B_I matches the three largest agendas almost exactly, but no single Δt reproduces all nine: two middle curves ($\alpha = 0.23$, 0.33) sit above ours, and the two largest agendas have a lower published B_A than we obtain at any Δt that fits the rest. Either the society size differs from ours or those middle curves, which overlap within a few pixels in the printed figure, cannot be digitised reliably; we cannot distinguish the two. Dropping the ambiguous readings does not move the selected value. All the Δt-independent checks pass: the above/below-diagonal ordering with its crossover at $\alpha \approx 1.7$, the sharp sign flip of $R_{\mu,c}$ at $d \approx 0$, and the $R_{c,w}$ wedge confined to large d and large f_d.

$N = 40$ is large enough. Repeating five representative points of the plane at $N \in \{20, 30, 40, 60\}$ with twelve realisations each, the order parameters at $N = 60$ differ from those at $N = 40$ by at most 0.02 ($R_{\mu,c}$, $R_{c,w}$), 0.03 (B_I) and 0.05 (B_A), while $N = 20$ differs by up to 0.24. The maps are therefore converged in N at $N = 40$, and the cost of going further is steep: total work grows as N^3.

Numerical safeguards. Three floors are applied, none of which the model prescribes: on the evidence Z, whose exact modulation functions diverge in the deep dissonance corners; on the affective variance V, which $F_V < 0$ drives monotonically towards zero; and on the covariance update, clipped to the boundary at which $C + a(C\hat{x})(C\hat{x})^\top$ ceases to be positive semi-definite. The last is the only one that can alter a trajectory, and it never triggered in the runs reported here.

I Protocol for the language-agent study

The interaction theatre. Every agent holds a private position on each issue of the agenda and a private disposition towards each other agent. A round draws an issue, an emitter and a receiver; the emitter states its position with a brief justification; the receiver sees the statement, the emitter's group label and its own current notes, and revises its position on that issue and its disposition towards that emitter. Rounds are drawn so that every ordered pair interacts comparably often, as in §3.

Instructions. Agents carrying the discriminating clause receive an otherwise identical prompt to those that do not. The clause instructs the agent to weigh credibility by group membership, at a ladder of strengths straddling zero so that the sharp transition at $d \approx 0$ can be resolved rather than merely bracketed. [fix the ladder and give the exact wording of every prompt]

Group labels and the swap control. Labels name arbitrary partitions, with no content relating to any issue on the agenda, so that any class-aligned structure is produced by the interaction rather than retrieved from the model's existing associations with real social categories. A control run swaps the labels between the two groups; the order parameters are invariant under that swap by construction, so a dependence on it would indicate label-driven priors rather than interaction-driven structure. [state the label scheme]

Elicitation. Opinion vectors and pairwise trust are elicited in separate passes that do not enter any agent's interaction history. Trust is elicited per ordered pair and never symmetrised: the affective balance B_A is defined on the two cycle orientations, and symmetrising destroys the distinction between organised hostility and frustration.

Reporting. [model version and decoding parameters, N, P, the number of rounds, the number of repeats per condition, and the raw elicited ρ and η matrices, released with the code.]